EXPERIMENT #1

Timestamp: 20260317_092329 | Environment: Local

1. Experiment Parameters & Network Statistics

Parameter	Value
Number of sticks (N)	30
Stick length (a)	0.7
Domain size	1.0 × 1.0
Source region	x < 0.1
Drain region	x > 0.9
Total intersections	67
Has percolation	✓ Yes
# Percolating paths	1
Nodes before pruning	67
Nodes after pruning	25
Edges before pruning	66
Edges after pruning	24

2. Network Visualizations

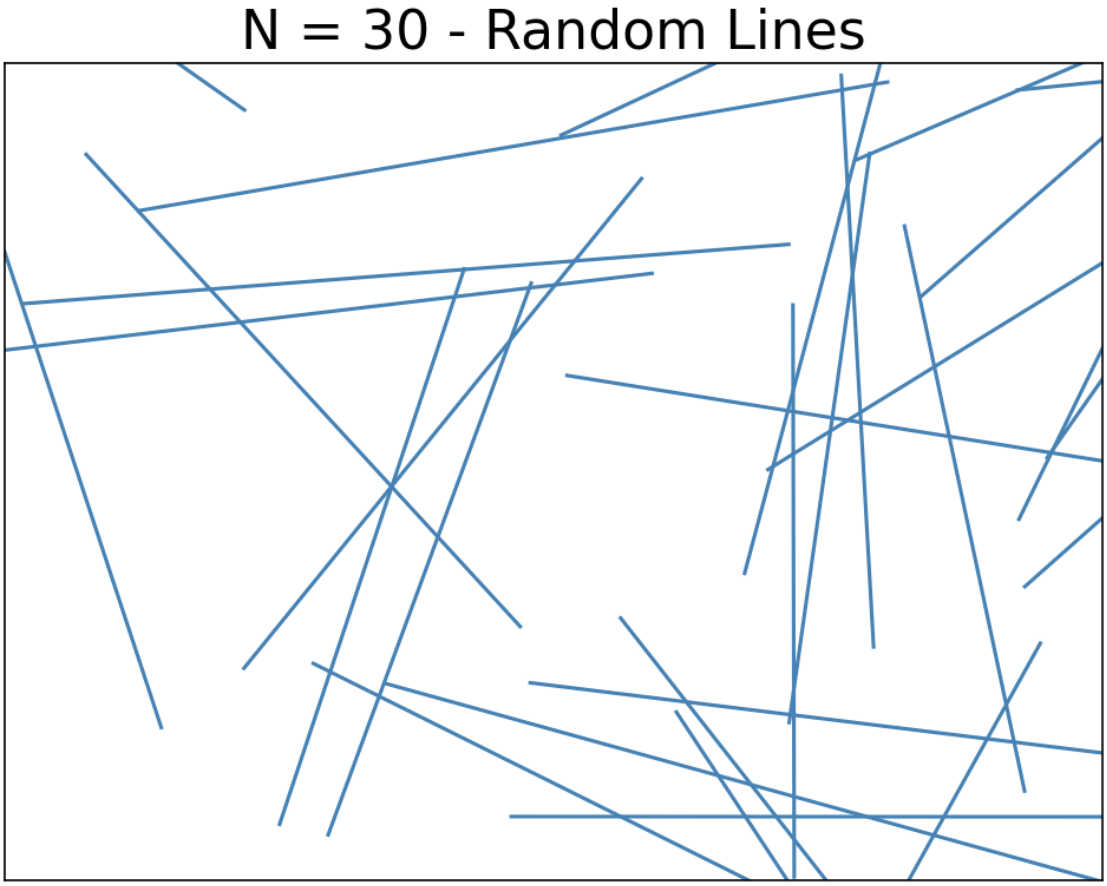


Fig. — 01 random lines N30

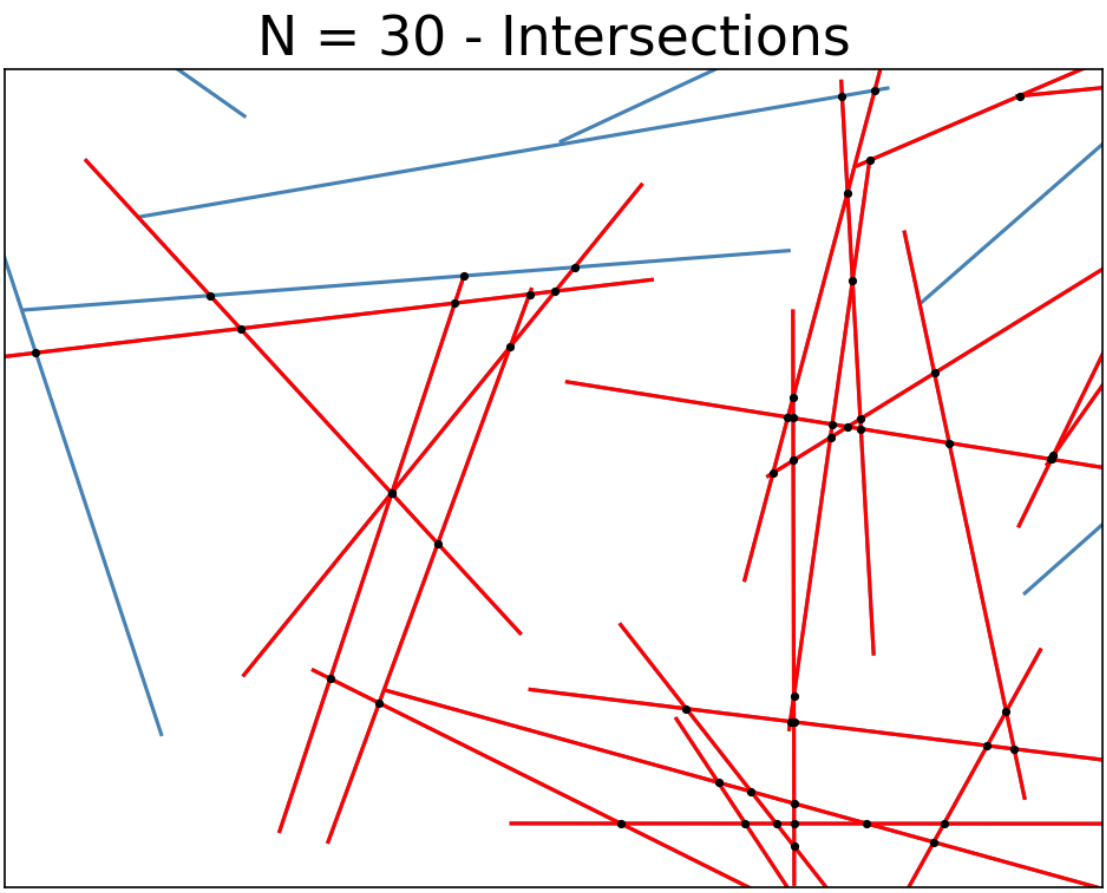


Fig. — 02 intersections N30

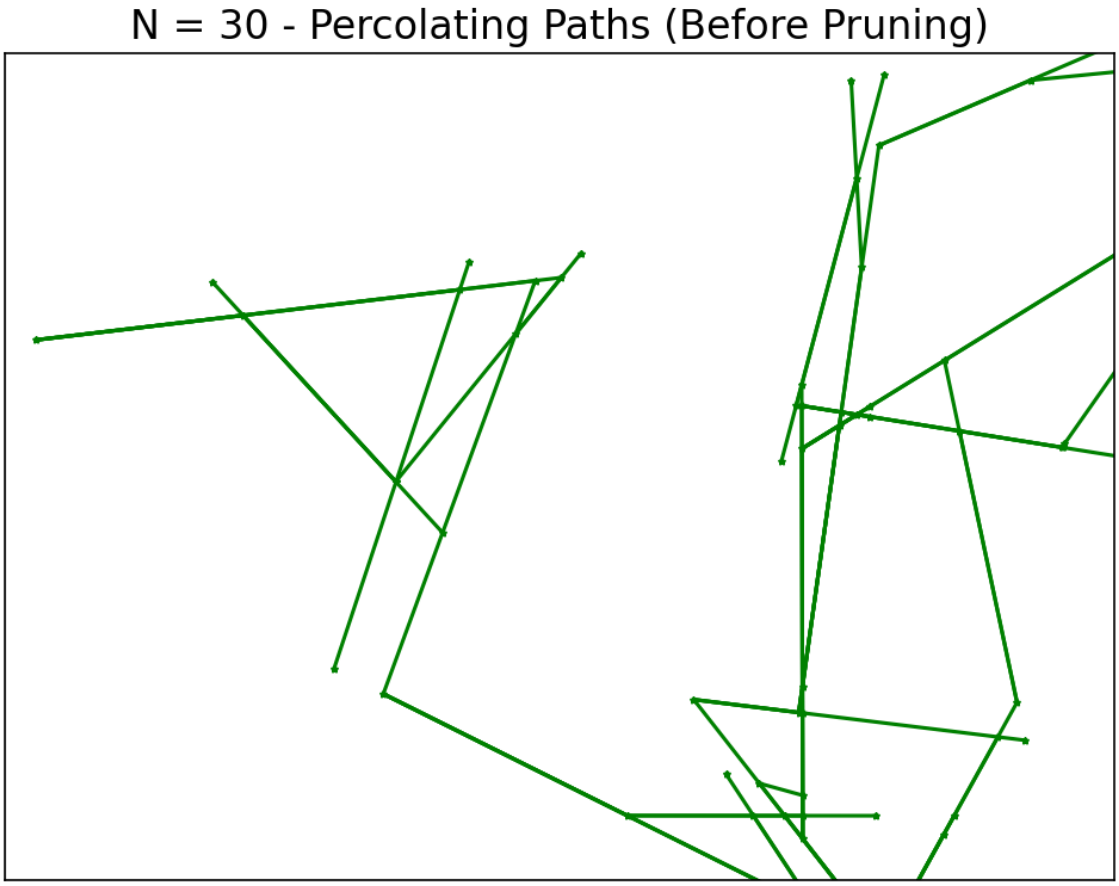


Fig. — 03 percolating paths before pruning N30

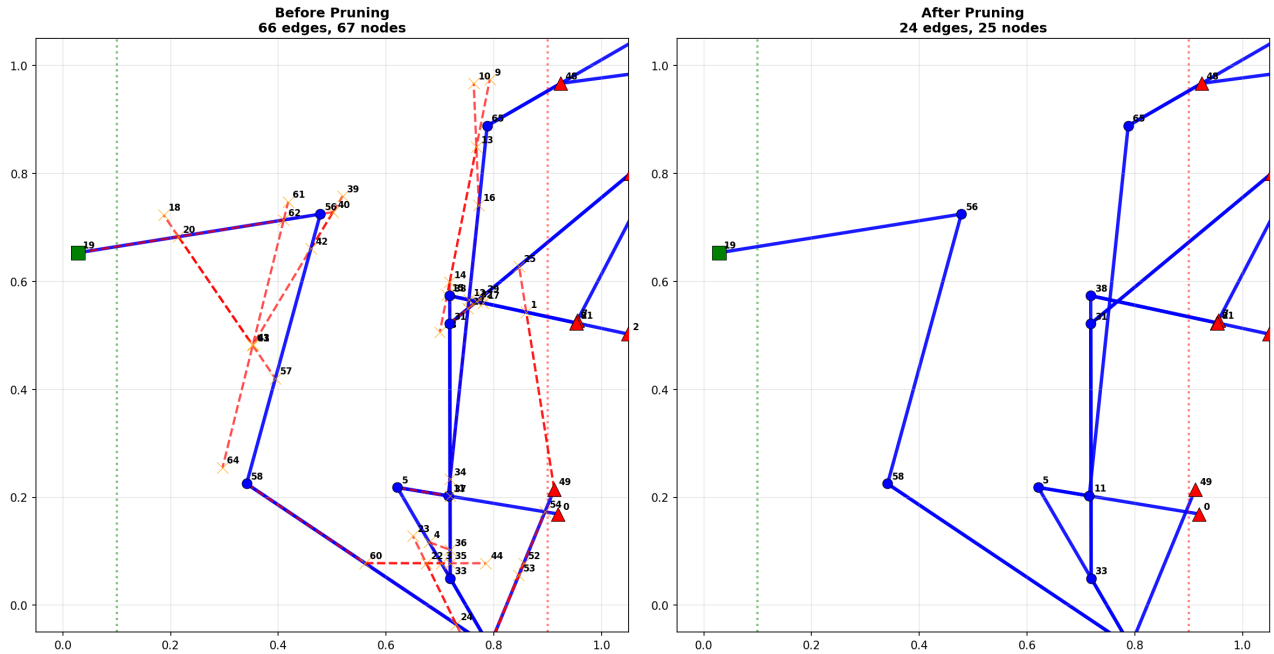


Fig. — 04 before after pruning N30

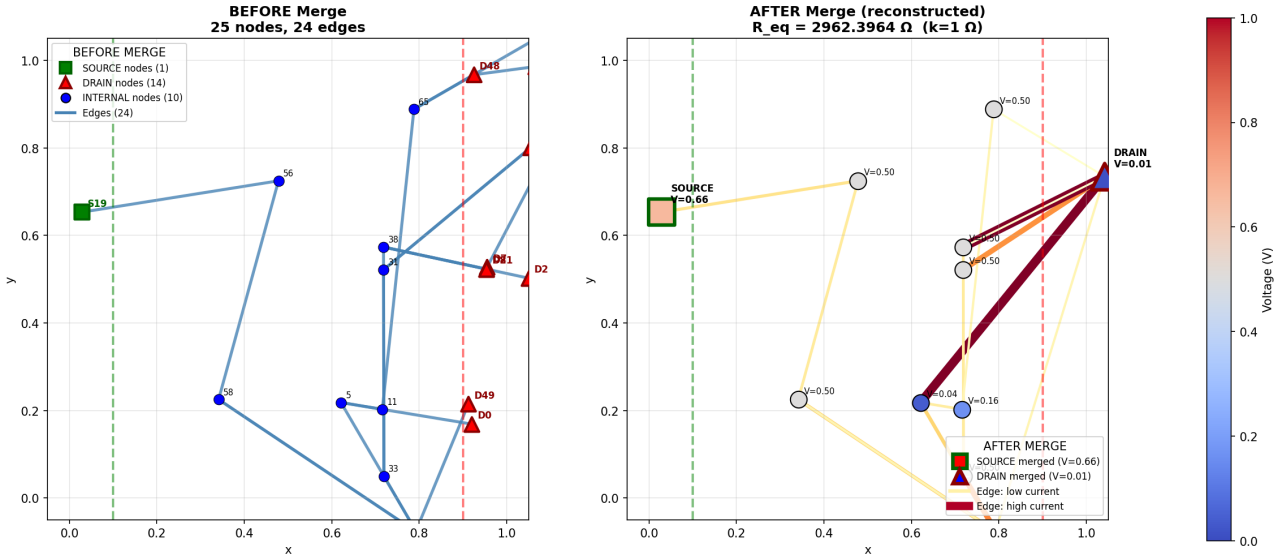


Fig. — 05 before after merge N30

3. Edge Resistances (Pruned Network)

Edge #	Node 1	Node 2	Conductance (S)	Resistance (Ω)	Voltage Drop (V)
0	19	56	1.0000	1.0000	0.000000
1	56	58	1.0000	1.0000	0.000000
2	58	6	1.0000	1.0000	0.162435
3	58	55	1.0000	1.0000	0.000000
4	6	5	1.0000	1.0000	0.146945
5	6	33	1.0000	1.0000	0.162435
6	55	49	1.0000	1.0000	0.000000
7	5	0	1.0000	1.0000	0.046902
8	5	11	1.0000	1.0000	0.015491
9	33	31	1.0000	1.0000	0.000000
10	33	38	1.0000	1.0000	0.000000
11	11	65	1.0000	1.0000	0.000000
12	31	26	1.0000	1.0000	0.000000
13	31	30	1.0000	1.0000	0.000000
14	38	2	1.0000	1.0000	0.037435
15	38	8	1.0000	1.0000	0.324871
16	38	21	1.0000	1.0000	0.000000
17	65	48	1.0000	1.0000	0.000000
18	65	66	1.0000	1.0000	0.000000
19	26	7	1.0000	1.0000	0.662435
20	26	45	1.0000	1.0000	0.000000
21	30	46	1.0000	1.0000	0.000000

Edge #	Node 1	Node 2	Conductance (S)	Resistance (Ω)	Voltage Drop (V)
22	30	59	1.0000	1.0000	0.000000
23	48	47	1.0000	1.0000	0.000000

4. Impedance Calculation

Quantity	Value
Applied voltage (V)	1.00 V
Resistance per segment k	1 Ω
Calculated impedance R_eq	2962.396431 Ω
Total current I	3.375645e-04 A

Impedance for different k (resistance per segment) values

k (Ω/segment)	R_equivalent (Ω)
1	2.962396
10	29.623964
100	296.239643
1000	2962.396431
10000	29623.964308